



Design concept of a high-altitude long-endurance unmanned aerial vehicle

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Abstract

This paper presents a conceptual project of a high-altitude long-endurance unmanned aerial vehicle. It describes a number of historical, current and prospective HALE aircraft and considers existing four serious obstacles, to overcome them can mean the successful building of HALE aircraft. Among these obstacles there are very special aerodynamic (low Reynolds numbers together with transonic speeds), very light structures usually of high aspect ratio, propulsion technology (usually propeller driven by turbocharged piston engines) and flight control system (usually combining the best features of preprogrammed and hand-flown modes). Four aerodynamic design concepts are presented and their performances are compared. Among them is a biplane, considered mainly because of its moderate wing span, which can be obtained for a relatively big wing area and a high effective wing aspect ratio, a relatively stiff wing structure, and lower induced drag being possible to be obtained at the same lift and wing area as for the equivalent monoplane. It is shown that an attentively designed biplane can be efficient aerodynamically for high altitude patrol missions having almost the same endurance, using the same fuel to reach a service ceiling and having the take-off mass (and the payload) considerably bigger than a corresponding, equivalent monoplane. The paper is completed with selected considerations about dynamic stability. © 1999 Elsevier Science Ltd. All rights reserved.

1. Introduction

Stratospheric flights can provide wide, new opportunities for human activity in various areas, as for example in commercial, scientific and military applications. Presumably the overall progress in

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the ultra high-altitude aerial vehicles technology is driven by military needs and was accelerated after the Persian Gulf War. However, it results in development of pure civilian missions such as weather forecast or investigations of stratospheric dynamics processes. The reason to fly at ultra high altitudes is not only to see further at wider angle, but also to be safer against manned interceptor aircraft and also to fly above most commercial aircraft. Among typical tasks there are early warning and ballistic missiles detection, observation and target identification, reconnaissance, forward air control and vectoring of air strikes, battlefield management, postengagement damage assessment, guidance and control including GPS navigation, border patrol, illegal air traffic control, sources of pollution emissions and transport of pollutants observation, crop and forest assessments, flood and fire control, monitoring of wildlife migration, volcanic eruptions and ocean currents, observation of highway traffic-patterns and congestion, and many others [1]. Although powered airplanes have reached ultra high altitudes (more than 20 km) and are able to operate continuously during a long time (several days), there are still such areas of flight envelope where challenges of high altitude and long endurance have not yet been met. After Epley [2] we will use the term high-altitude long-endurance (HALE) to describe the mission, which is typically a multiday surveillance that has to be performed at ultra high altitude. Such aircraft must be unmanned and operated autonomously or under remote control. Because of the required mobility and mission flexibility the balloons, widely used in the past, have to be replaced by faster and more mobile aircraft. U-2 and its derivatives are usually too expensive in civil operation, first of all because of their costly power unit, big size and many apparatus necessary for the aircraft to be operated by a human pilot. The recent technology advances [3] in instrumentation, sensors, actuators, electronics and power units give us the promise that HALE missions using small and inexpensive remotely controlled aircraft become the standard way of stratosphere exploration in the 21st century.

Selected parameters of a number of historical, current and prospective HALE aircraft (including conceptual projects) are shown in Fig. 1. These and many other stratospheric aircraft are presented and detailly described in numerous references [2–7]. However, it should be emphasized that hundreds of UAV programs (including HALE missions designs) ended without becoming practical. It seems that there exist at least four serious obstacles – to overcome them can mean the successful building of HALE aircraft. These obstacles are very special aerodynamic conditions (low Reynolds Numbers together with transonic speeds), very light structures usually of high aspect ratio, propulsion technology (usually propeller driven by turbocharged piston engines) and flight control system (usually combining the best features of preprogrammed and hand-flown modes). In our times still the degree of automation is technology driven and some difficult to automate phases of flight (take-off for example) are piloted by human. Aerospace experts expect that rapid progress in computer technology, sensors, materials, power units and human interface technology make it possible soon that UAV systems will have ‘almost infinite levels of automation and human control’ [8]. For example, in 1997 the solar-powered UAV, Pathfinder broke the high-altitude record for the highest flying propeller-powered aircraft, reaching 21.450 km [9]. In planning the mission of HALE UAV selected meteorological data should be taken into account. The use of climatology, surface and upper atmospheric observations, weather forecast and real-time weather data are discussed in [10], especially for Pathfinder solar-powered UAV. In the present article the atmosphere is assumed to be calm, not windy and consistent with the International Standard Atmosphere.

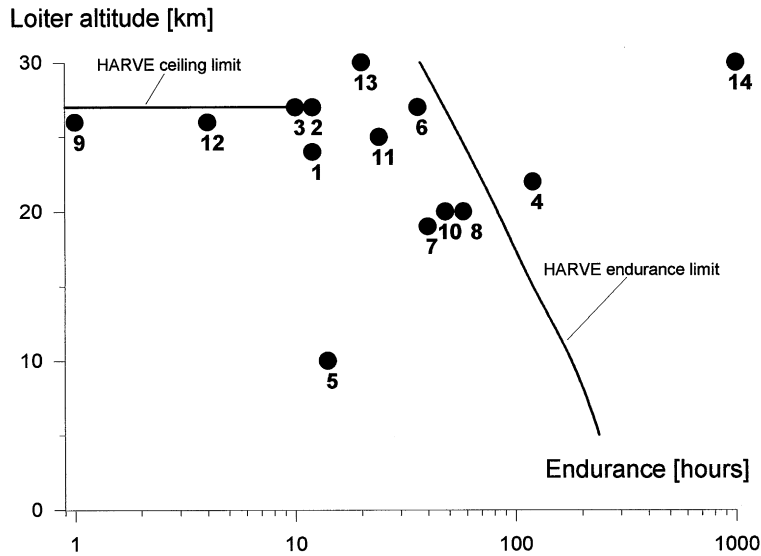


Fig. 1. Endurance and loiter altitude for selected stratospheric aircraft: (1) U2; (2) ER-2 (Earth Resources); (3) Strato 2C; (4) Pathfinder; (5) Tier III—DarkStar; (6) Theseus; (7) Tier II + Global Hawk; (8) Condor; (9) Perseus A; (10) Perseus B; (11) Altus; (12) Alliance; (13) Centurion; (14) Helios.

This paper is devoted mainly to selecting the main design parameters of HALE UAV, which fulfil the prescribed mission. It considers also some aspects of aerodynamics, propulsion and maintenance problems.

2. Prescribed mission and payload

The main task of the aircraft under consideration is reconnaissance and remote sensing missions. It will be an unmanned aerial vehicle, flying at ultra high altitude (up to 27 km) and long endurance (up to 48 h) (HALE UAV). It should be inexpensive in initial and operating costs. To be expected are minimal maintenance and fast turnaround. Construction should be modular for easy transportation. The basic option is payload weight of 300 kg but depending on expected endurance the payload weight is flexible in the range from 100 to 1000 kg. Structure based on glider, fibre-glass/carbon technology should be extremely light. Also a low-cost fabrication and assembly is expected. The main design parameters are as follows:

- departure altitude (maximum) 27 km
- operational speed 100 m/s
- endurance 48 h
- payload weight (optional) 300 kg
- payload weight (flexible) 100–1000 kg

A typical mission profile is shown in Fig. 2.

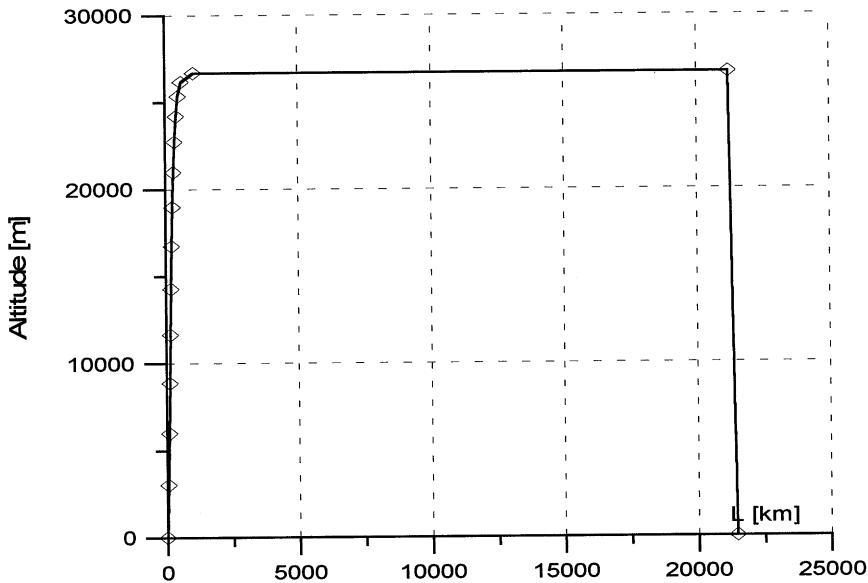


Fig. 2. Profile of assumed mission—altitude versus horizontal distance.

3. Power unit

Existing, general utility powerplants do not have the level of performance needed for HALE UAV. If UAV has to be inexpensive in operation, then airbreathing engines are the only possible near-term option. A short review of existing engines used in the past on different aircraft (on the U-2 for example) and now (on Strato 2C for example) is given in [2].

Sufficient power at departure altitude could be obtained only by the use of a multistage turbocharger with a liquid intercooler. The classical spark ignition engine should power a propeller of big diameter through a gearbox of variable reduction ratio. Three different specific solutions are considered:

- using the existing powerplant of very similar purpose and available power. It could be for example the engine used to power the Strato 2C aircraft, based on Teledyne Continental 550 core engine, designed to maintain a constant power of 300 kW to an altitude of 24 km (see Fig. 5 in [2]),
- creating a similar powerplant of higher power (400 kW), which could be based on the engine Allied Signal – Lycoming IO-720. This engine does not fulfil the specific requirement at the moment, but because of its big cubic capacity, could be a good choice with a power excess after necessary modification. However, such modification is a serious task and should be done parallel to the structure design programme. The scheme of the above-mentioned power unit is given in Fig. 3,
- creating the powerplant of 400 kW, based on the stratified charge rotary (Wankel) engine of the 170 series made by Rotary Power International. This solution could be advantageous because of its low weight and a number of different fuels which could be used. However, this engine cannot be considered as a basic for current use, rather as a propulsion system in the future.

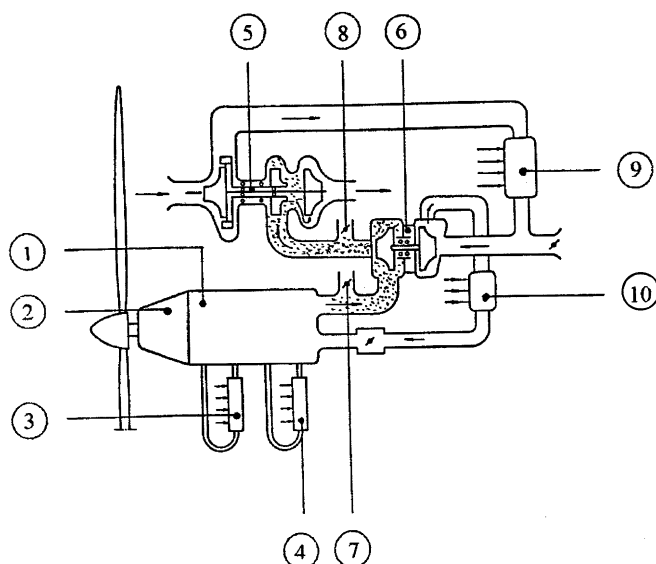


Fig. 3. Scheme of power unit: (1) Engine—Lycoming 10-720, (2) Speed reducer, (3) Liquid cooler, (4) Oil cooler, (5) Rotary compressor—first stage, (6) Rotary compressor—second stage, (7) Waste gas bleeding 1, (8) Waste gas bleeding 2, (9) compressed-air-cooler 1, (10) compressed-air-cooler 2.

In the two last cases a special propeller was selected from the family of Avia-Hamilton Ltd. As a turbocharger the bi-rotary compressor, developed in Institute of Aviation (Poland) is planned to be used [11]. The altitude performance of an engine, used for aircraft performance simulation, is shown in Fig. 4.

None of the power plants from three cases mentioned above is satisfactory: the first does not deliver the 400 kW power required, the second must be modified according to Fig. 3 and the third is only a future option. However, because the modifications (case II) and especially the new design strategy (case III) are long, risky processes, the first solution (i.e. the power plant used for Strato 2C) has been selected. It will limit the ceiling below 27 km but enables to develop the design just now.

4. Low-Reynolds and high-Mach numbers range of flight

Modern composite structural technology allows a rather low wing-loading of 200 N/m² and a moderately high aspect ratio—less than 25. Following Drela [12] in the further analysis we will use the so-called reduced Reynolds number R and reduced Mach number M according to the formulae

$$R = Re \sqrt{C_L}, \quad M = Ma \sqrt{C_L}. \quad (1)$$

It can be shown that reduced Reynolds and Mach numbers do not depend on trim changes, i.e. as speed and lift coefficient vary at the same altitude and weight then R and M do not change.

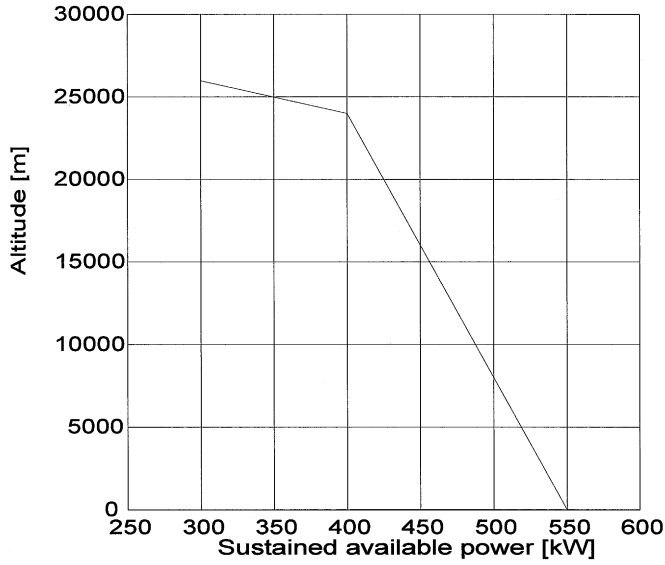


Fig. 4. Available power versus altitude.

We have

$$R = \text{Re} \sqrt{C_L} = \frac{\rho V c}{\mu} \left(\frac{2W}{\rho V^2 S} \right)^{1/2} = \left(\frac{2\rho}{\mu^2} \frac{W}{A} \right)^{1/2}, \quad (2)$$

$$M = \text{Ma} \sqrt{C_L} = \frac{V}{a} \left(\frac{2W}{\rho V^2 S} \right)^{1/2} = \left(\frac{2}{\gamma p} \frac{W}{S} \right)^{1/2}, \quad (3)$$

and from above it is clear that at fixed altitude and weight the reduced Reynolds and Mach numbers remain constant, whereas both Re and Ma change, i.e.

$$\text{Re} = \frac{R}{\sqrt{C_L}}, \quad \text{Ma} = \frac{M}{\sqrt{C_L}}, \quad (4)$$

where ρ denotes density, μ the viscosity, γ the specific heats ratio, and p the ambient pressure.

In [12] it is shown that for limiting $W/S = 200 \text{ N/m}^2$, $A = 25$ and assuming $S = 70 \text{ m}^2$, altitudes of up to 35 km can be reached theoretically. Above these altitudes the reduced Mach number is greater than 0.7 and the reduced Reynolds number is less than 200,000. This high-Mach and low-Reynolds number combination (see Fig. 5) is quite unique in aeronautical engineering and almost not investigated in the past. It is below the critical Reynolds number for most of airfoils and above the critical Mach number. Knowing that low-Reynolds number airfoils tend to have their best efficiency close to the stall and trying to reduce the Mach number to decrease the wave drag it is desirable to employ the largest feasible lift coefficient [12–14].

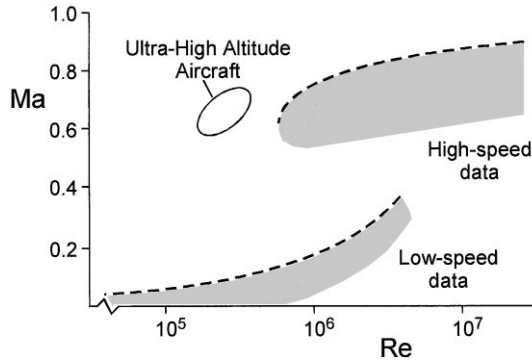


Fig. 5. Limits of available airfoil data (after [12]).

5. Selection of preliminary design

The objective is to arrive at a reasonably detailed layout of an airplane configuration. The detailed goal is to ‘fine tune’ [15] a configuration that indeed meets all important requirements laid down in the patrol mission specification. In selecting the configuration for high-altitude long-endurance mission the following assumptions have been taken into account:

- the key element for the whole aircraft is its wing. Because of the extremely low air density, high weight of aircraft (due to large fuel capacity) and moderate speed of flight, a sufficiently big gross wing area is needed. The configuration being able to minimise the wing unit mass, is a biplane [16,17];
- one of the well-known disadvantages of a biplane configuration is its increased induced drag. This induced drag can be minimised due to:
 - separation of a part of lifting surface into a lifting horizontal tail (the third wing);
 - selecting the aspect ratio as high as possible;
 - selecting the ‘best’ values of gap, stagger and decalage of the wing cellule [18];
 - the airfoil should have a very high design lift coefficient. A value of $C_L = 1.2$ (or higher) is to be desired. To minimise the wing structure weight, the airfoil should be thick, with a low pitching moment value. This airfoil at low-Reynolds number should possess the high efficiency (measured by a good C_L/C_D ratio). The operational Mach number should be below the drag-critical Mach number. A number of numerical results, based on the coupled viscous/inviscid ISES code (a conservative finite-volume streamline grid Euler formulation) [13,14] have been presented in [12] for the Eppler 387 airfoil, for a circular arc airfoil and for the TH-family airfoils. For the project under consideration the Wortmann-FX 67-K-170/17 wing section was used.

The biplane was considered mainly because of the following reasons [16,18–20]:

1. a moderate wing span can be obtained for a relatively big wing area and a high effective wing aspect ratio;
2. a relatively stiff wing structure can be built for a big wing area and moderate structural mass;

3. for a given structural mass the moments of inertia (in roll and yaw) can be relatively small;
4. it usually does not need costly high lift devices;
5. it can be aerodynamically more efficient (lower drag at the same lift and wing area).

All above-mentioned factors cause that a biplane configuration is regarded as the good choice for patrol mission HALE airplane. Also, the stiff structure, high effective aspect ratio, big fuel mass and very big wing area at limited wing span are of primary interest. Attentively designed biplane can be more efficient aerodynamically than corresponding, equivalent monoplanes. An excellent comparison between both configurations, assuming that they have the same lift capacity and hence wing areas, is given in [18]. Both configurations have had ‘similar’ induced drag characteristics, namely monoplane: $b = 0.77$ m, $c = 0.166$ m, $A = 4.615$, $e = 0.92$, $A_e = 4.24$ and biplane: $b = 0.627$ m, $c = 0.102$ m, $A = 3.0625$, $\sigma = 0.55$, $e = 0.775$, $A_e = 3.96$, $\text{gap}/b = 0.16$, $\text{stagger}/c = 0.87$, $\text{decalage} = -6^\circ$. The profile drag is assumed to be the same for both biplane and monoplane configurations. From wind tunnel experiments made by Olson [18], it follows that for the same lift C_L , drag coefficient C_D can be reduced even by 25% and L/D ratio can be increased by 16%. This increase in biplane efficiency is caused due to the changed flow distribution and hence the pressure distribution over the lifting surfaces. Biplane endurance—parameter being very important for patrol aircraft—can be even increased on several percents comparing to the-same-lift-capacity-monoplane.

Together with a biplane configuration a number of other designs have been considered [1,11,25]. The common feature of these variants is the large diameter propeller, rotating with low angular velocity due to the gearbox (reduction ratio is in the range from 2.5 to 4, depending on configuration). So, in the case of the flying wing, considered together with some biplanes and extremely high wing monoplane, a special, variable setting of propeller-axis has to be applied in order to keep a necessary propeller clearance during take-off and landing. The most representative examples being considered, are shown in Fig. 8. They are:

1. forward swept flying wing. It was considered with the hope of its efficiency, low induced drag, low form drag and a good bending moment distribution along wing span,
2. high-wing monoplane,
3. twin boom biplane with butterfly tail,
4. twin boom biplane with lifting tail.

The wing size for flying wing and both biplanes has been selected by using the assumption that the fuel tank and wing fuel compartment capacity is equal to 2000 kg. For the high-wing monoplane this capacity was assumed to be 1500 kg. Wing weight in all cases was estimated basing on statistical analysis.

Important aerodynamic characteristics—lift, drag, pitching moment, downwash and the horizontal tail lift curve slope versus angle of attack and elevator deflection—have been computed for different altitude, speed and equilibrium conditions, either by panel methods [21–23] or simple engineering methods [24]. Some of such characteristics are shown in Figs. 6 and 7.

Fig. 6 shows three computed polars in which the reduced parameters R and M are held fixed. Fig. 7 shows lift and drag coefficients at different altitudes. For all curves in Fig. 7 the reduced parameters R and M change due to changing of weight (see Eqs. (2) and (3))—the mass of the flying wing configuration decreases from 4500 to 2200 kg. The drag-increase at constant lift, or the

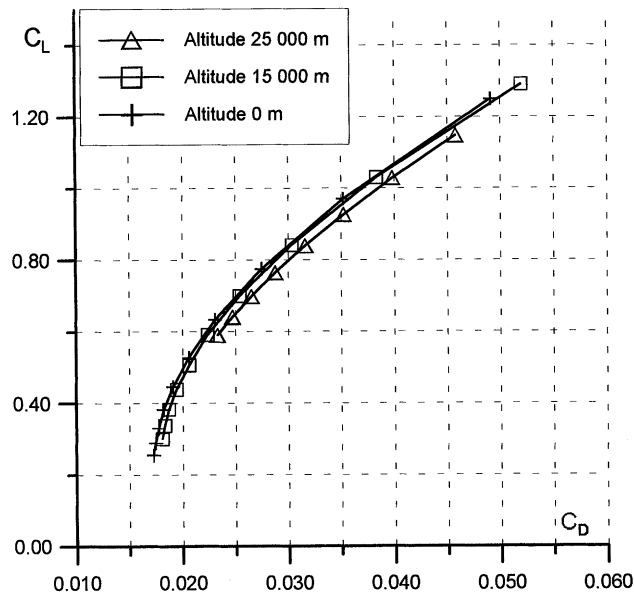


Fig. 6. Polars of equilibrium at three different altitudes (flying wing configuration).

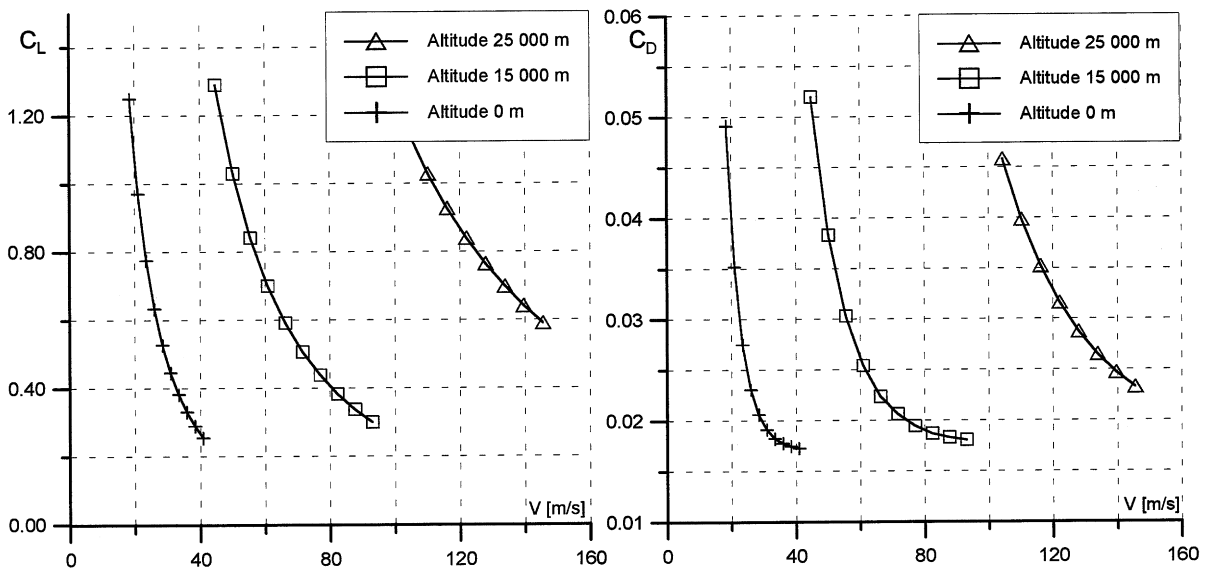


Fig. 7. Lift and drag coefficients versus flight speed at different altitudes and aircraft mass (flying wing configuration).

drag-increase at constant speed, both versus altitude-increase can be observed in Figs. 6 and 7. Also, a lift-increase at constant speed versus altitude-increase can be observed at Fig. 7. The higher lift and drag coefficients are due to differences in air density (not due to compressibility which was not taken into account in these considerations).

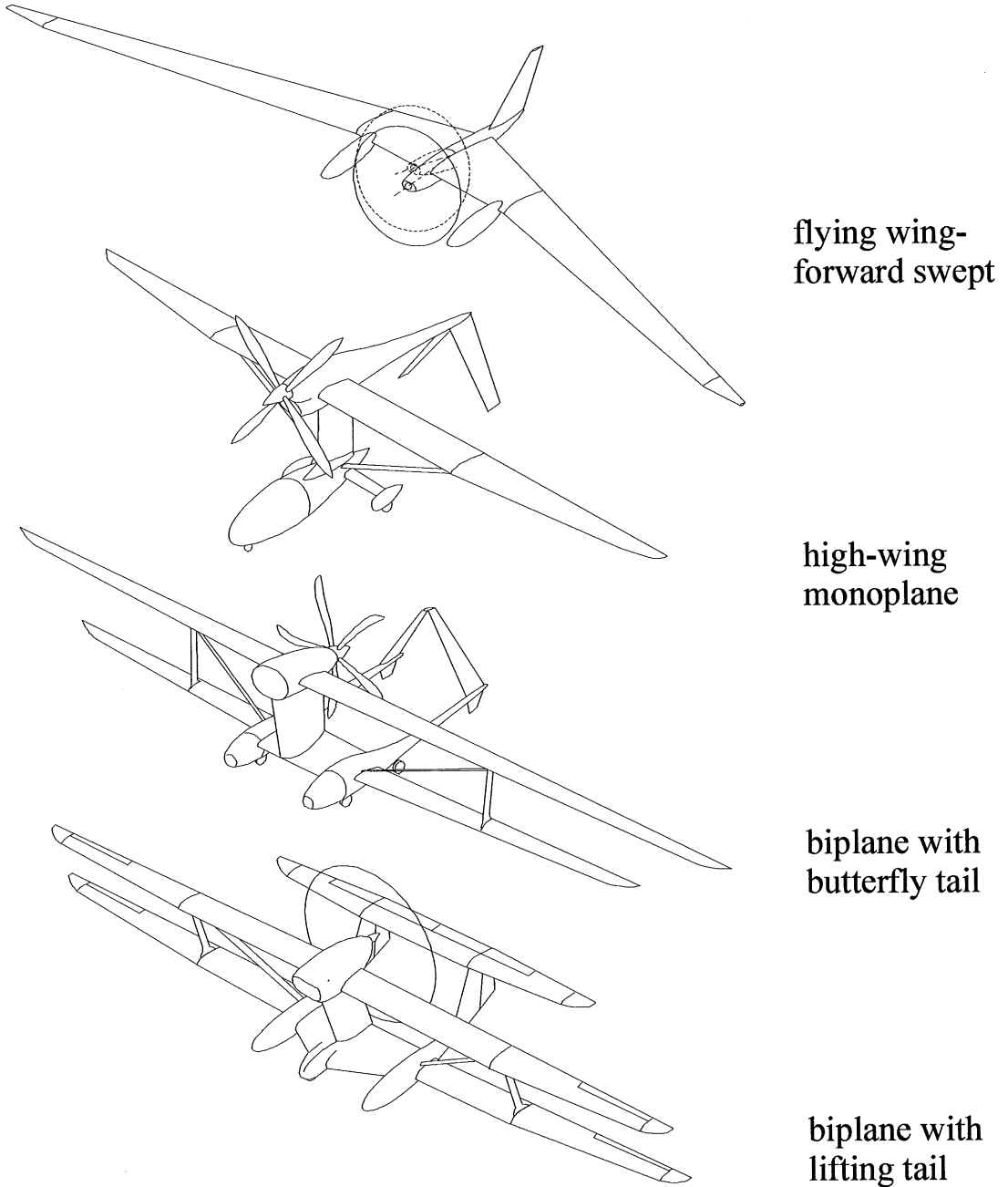
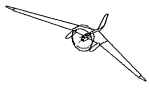
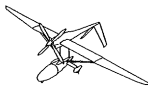
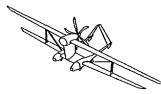
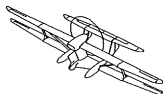


Fig. 8. Configurations considered.

Table 1
Selected performances computed for considered configurations

Configuration	Flying wing-forward swept	High-wing monoplane	Biplane with butterfly tail	Biplane with lifting tail
				
Take-off mass (kg)	4200	3260 ^a	3550	3550
Operating empty mass/payload/fuel mass (kg)	1900/300/2000	960/300/2000 ^a	1250/300/2000	1250/300/2000
Wing section	FX 67K 170/17	FX 67K 170/17	FX 67K 170/17	FX 67K 170/17
MAC/span (m)	3.56/50	1.76/34	1.4/32	1.4/32
Effective Aspect Ratio	14.75	19.27	13	15.5
Zero-lift drag coefficient	0.010	0.013	0.019	0.016
Wing area (m ²)	169	60	115	115
Absolute ceiling (km)	27.15	26.91	26.2	26.95
Service ceiling (km)	26.58	26.46	25.77	26.49
Endurance at service ceiling (h)	48.3	54.0	46.8	51.3
Speed for maximum endurance at ceiling (m/s)	96.1	134.8	97.8	103.8
Range at service ceiling (km)	18,400	26,990	17,100	19,700
Speed for maximum range at ceiling (m/s)	114.1	143.5	109.1	113.9
Fuel to reach altitude 25 km (kg)	221	182	210	187.4
Endurance at altitude 25 km (h)	54.5	58.2	49.6	56.8
Speed for maximum endurance at 25 km (m/s)	85.1	121.0	92.4	93.0
Range at altitude 25 km (km)	19,300	27,000	17,400	20,200
Speed for maximum range at altitude 25 km (m/s)	109.0	142.4	107.3	110.1
Maximum rate of climb at sea level (m/s)	1.6	2.2	1.7	1.9
Maximum rate of climb (m/s)	4.6	5.8	5.1	5.4
Altitude where rate of climb is maximum (km)	14.6	14.4	14.5	14.8
Time to reach 25 km altitude (min)	139	115	132	119
Time to reach service ceiling (min)	164	137	146	141
Minimum/maximum speed at $h = 0$ (m/s)	14.7/41.0	21.8/53.7	16.4/39.5	16.4/42.0
Specific endurance at altitude 25 km (min/kg)	1.63	1.75	1.49	1.70
Specific range at altitude 25 km (km/kg)	9.65	13.5	8.7	10.1

^aCapacity of integral fuel tank and wing fuel compartments in high-wing monoplane configuration is limited to 1500 kg. However, to compare this configuration performances with that of other configurations the capacity of the high-wing configuration was increased (virtually) to 2000 kg. In fact, the endurance and range at 25 km altitude for high-wing monoplane configuration will be reduced.

Looking at Table 1 we see that the high-wing monoplane has the best specific endurance (ratio of endurance to the total fuel mass) and the best range (ratio of range to the total fuel mass). Also, this configuration has the highest rate of climb and the shortest time to reach service ceiling. Its high speeds in patrol, high altitude flight (135 m/s at service ceiling, corresponding to a Mach number equal to 0.45) are however below the critical Mach number. High speeds could be prohibited for the biplanes, especially flying with high lift coefficient and at high angles of attack, not for the monoplane. The high-wing monoplane can fly at higher Mach number and lift coefficient.

Selected performance parameters for the flying wing configuration are shown in Fig. 9 under the assumption that total fuel mass was 2300 kg. All other parameters are consistent with that of presented in Table 1.

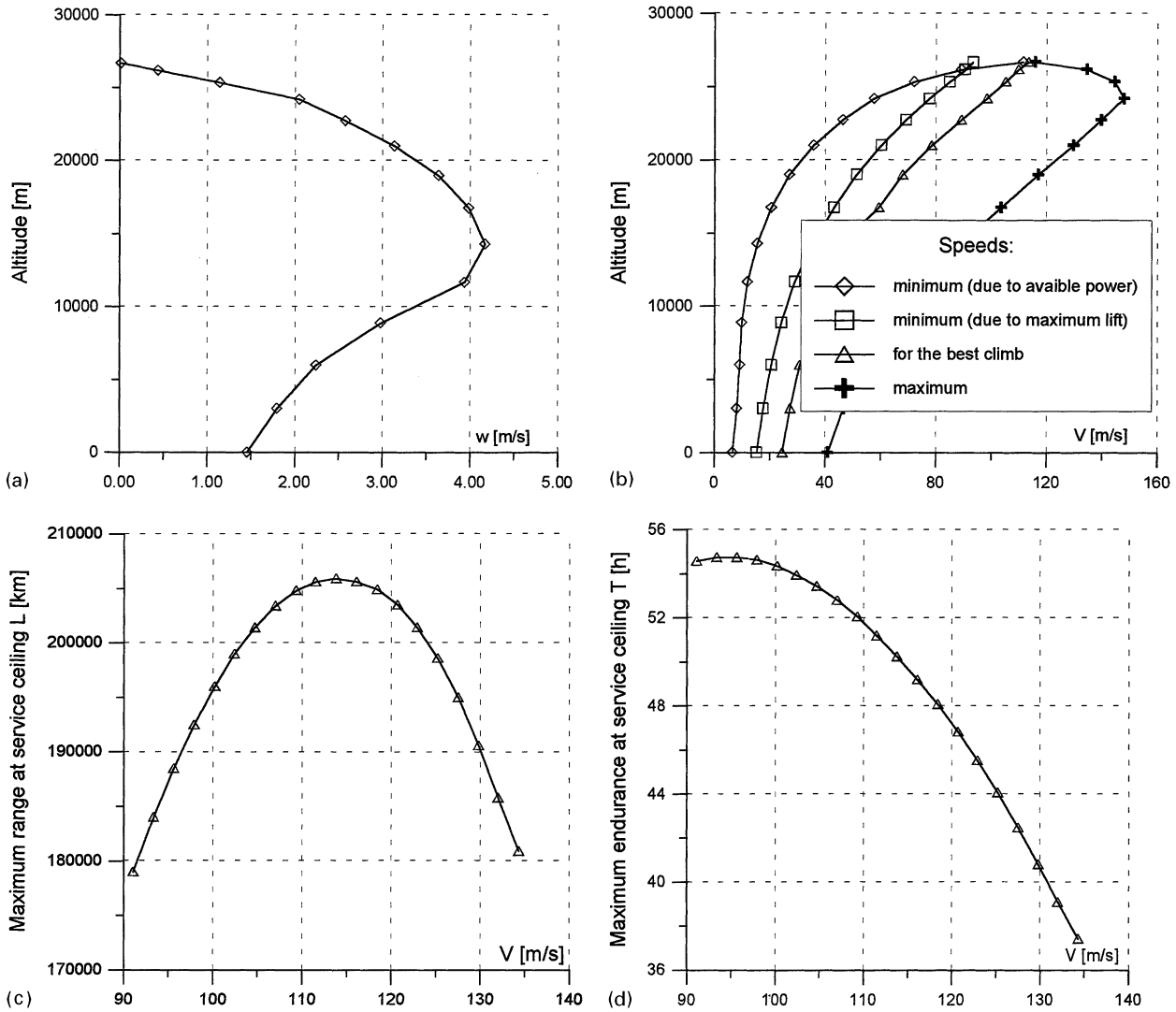


Fig. 9. Performance characteristics of the flying wing configuration. (a) Maximum rate of climb versus altitude; (b) Speeds versus altitude; (c) Maximum range at ceiling versus flight speed; (d) Maximum endurance at ceiling versus flight speed; (e) Fuel consumption (m) and time-to-climb (time) versus altitude.

Table 2 compares the performances of the same aerodynamic configuration (flying wing) for different take-off masses (fuel mass was assumed to be 1000, 1500, 2000 and 2300 kg, respectively). Most of performances vary almost linearly against initial mass of fuel. Increasing the fuel to double results in doubling the range and endurance and in decreasing the rate of climb, by approximately 30%. It should be emphasised that the longest endurance and range correspond to different speeds. However, in both cases the speed for longest endurance (or longest range) was assumed to be constant during the computing cycle. In fact, it is true only approximately (see Figs. 10 and 11), so both these performance parameters could be increased.

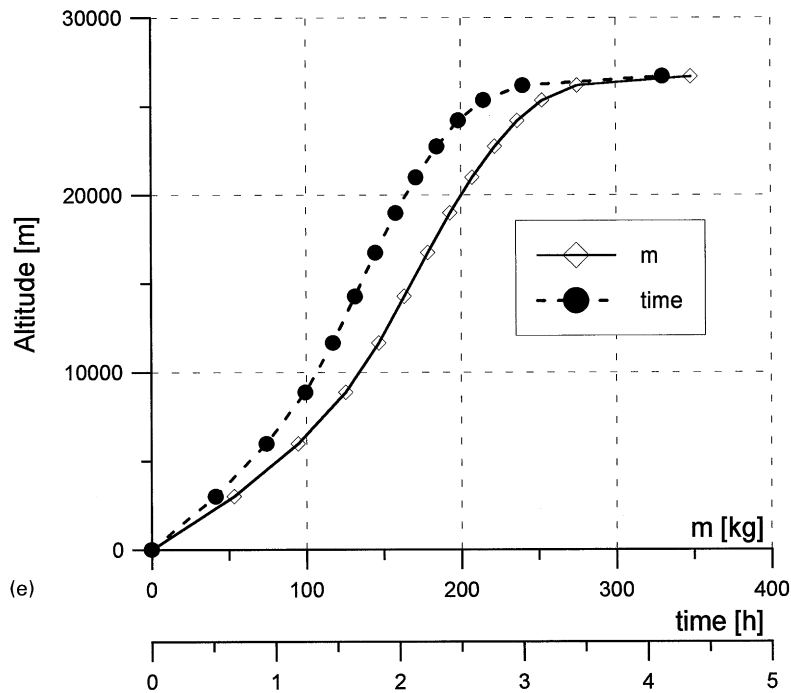
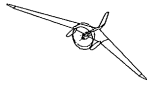
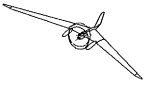
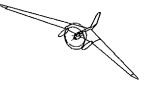
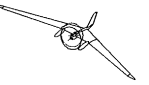


Fig. 9. Continued

Table 2
Selected performances computed for flying wing versus fuel mass

Configuration	Flying wing- forward swept	Flying wing- forward swept	Flying wing- forward swept	Flying wing- forward swept
				
Fuel Mass	1000	1500	2000	2300
Take-off mass (kg)	3200	3700	4200	4500
Absolute ceiling (km)	29.01	28.06	27.15	26.69
Service ceiling (km)	28.49	27.51	26.58	26.10
Endurance at service ceiling (h)	23.4	36.3	48.3	54.8
Speed for maximum endurance at ceiling (m/s)	101.9	99.4	96.1	95.6
Range at service ceiling (km)	9,300	14,200	18,400	20,600
Speed for maximum range at ceiling (m/s)	119.0	117.0	114.1	113.9
Fuel to reach altitude 25 km (kg)	150	183	221	248
Endurance at altitude 25 km (h)	30.7	43.8	54.5	59.5
Speed for maximum endurance at 25 km (m/s)	80.7	82.0	85.1	87.4
Range at altitude 25 km (km)	10,300	15,200	19,300	21,300
Speed for maximum range at altitude 25 km (m/s)	106.0	111.0	109.0	109.4
Maximum rate of climb at sea level (m/s)	2.1	1.7	1.6	1.5
Maximum rate of climb (m/s)	6.6	5.4	4.6	4.2
Altitude where rate of climb is maximum (km)	12.9	15.1	14.6	14.3
Time to reach 25 km altitude (min)	94	115	139	157
Time to reach service ceiling (min)	130	147	164	177
Minimum/maximum speed at $h = 0$ (m/s)	12.8/41.1	13.8/41.0	14.7/41.0	15.2/40.9

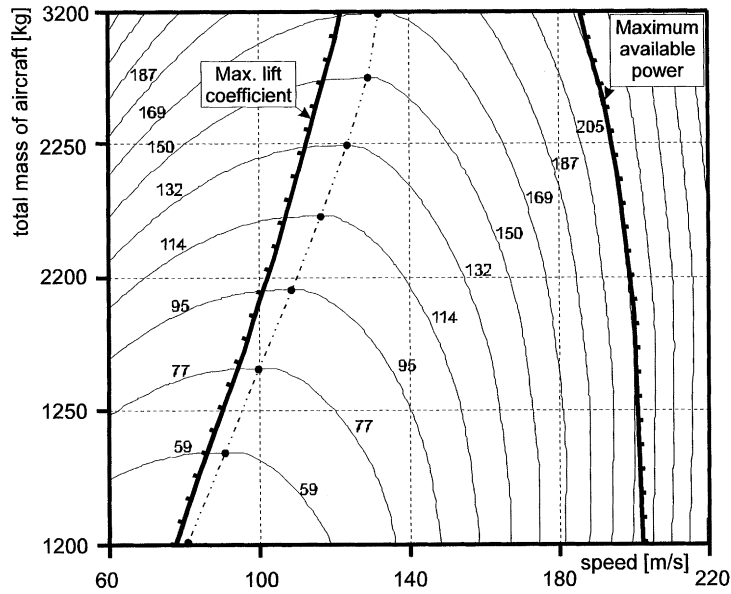


Fig. 10. Power (in kW) required to fly at 25 km altitude for the high-wing monoplane.

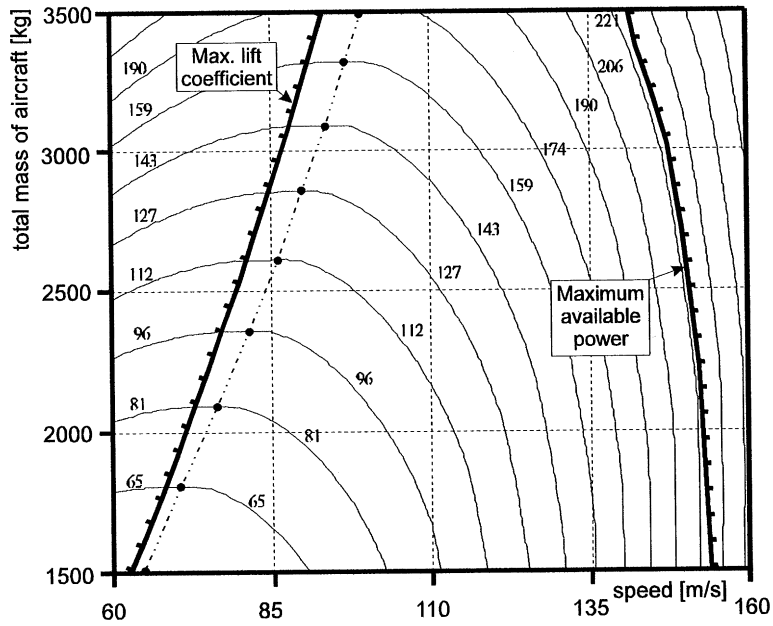


Fig. 11. Power (in kW) required to fly at 25 km altitude for the biplane with lifting tail.

Table 3
Selected performances computed for biplanes of different aspect ratio

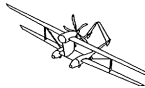
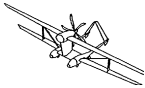
Configuration	Biplane with butterfly tail $A_e = 13$	Biplane with butterfly tail $A_e = 10$
		
Take-off mass (kg)	3550	3200
Operating empty mass/payload/fuel mass (kg)	1250/300/2000	900/300/2000
Wing section	FX 67K 170/17	FX 67K 170/17
Equivalent monoplane Aspect Ratio	13	10
Zero-lift drag coefficient	0.019	0.019
Wing area (m ²)	115	115
Absolute ceiling (km)	26.2	26.1
Service ceiling (km)	25.77	25.70
Endurance at service ceiling (h)	46.8	50.0
Speed for maximum endurance at ceiling (m/s)	97.8	92.5
Range at service ceiling (km)	17,100	17,700
Speed for maximum range at ceiling (m/s)	109.1	105.6
Fuel to reach altitude 25 km (kg)	210	192
Endurance at altitude 25 km (h)	49.6	52.4
Speed for maximum endurance at 25 km (m/s)	92.4	87.9
Range at altitude 25 km (km)	17,400	17,870
Speed for maximum range at altitude 25 km (m/s)	107.3	104.6
Maximum rate of climb at sea level (m/s)	1.7	1.9
Maximum rate of climb (m/s)	5.1	5.6
Altitude where rate of climb is maximum (km)	14.5	14.3
Time to reach 25 km altitude (min)	132	121
Time to reach service ceiling (min)	146	135
Minimum/maximum speed at $h = 0$ (m/s)	16.4/39.5	15.3/39.2
Specific endurance at altitude 25 km (min/kg)	1.49	1.57
Specific range at altitude 25 km (km/kg)	8.7	8.93

Table 3 presents a comparison between configurations of the same layout (biplane with butterfly tail), for the same wing area but different aspect ratio. It shows that a decreasing aspect ratio (and keeping the same zero-lift drag coefficient) increases endurance and range. It is due to fact that the lower aspect ratio saves the structure mass and, independent of penalties in aerodynamics, improves most performance parameters.

Isolines shown in Figs. 10 and 11 correspond to the required power curves (kW) being constant. The horizontal axis defines the flight speed, the vertical axis defines the aircraft mass—variable during flight. Two bold denticulated lines correspond to limits of flight parameters—the left one to the stalling lift coefficient and the right one to insufficient available power. The thin dashed line connects the points of minimum required power curves—corresponding to the different aircraft mass. The thin dashed line corresponds to the best flight parameters—giving the longest endurance. Fig. 10 was obtained for the high-wing monoplane configuration, Fig. 11 for the biplane with lifting tail.

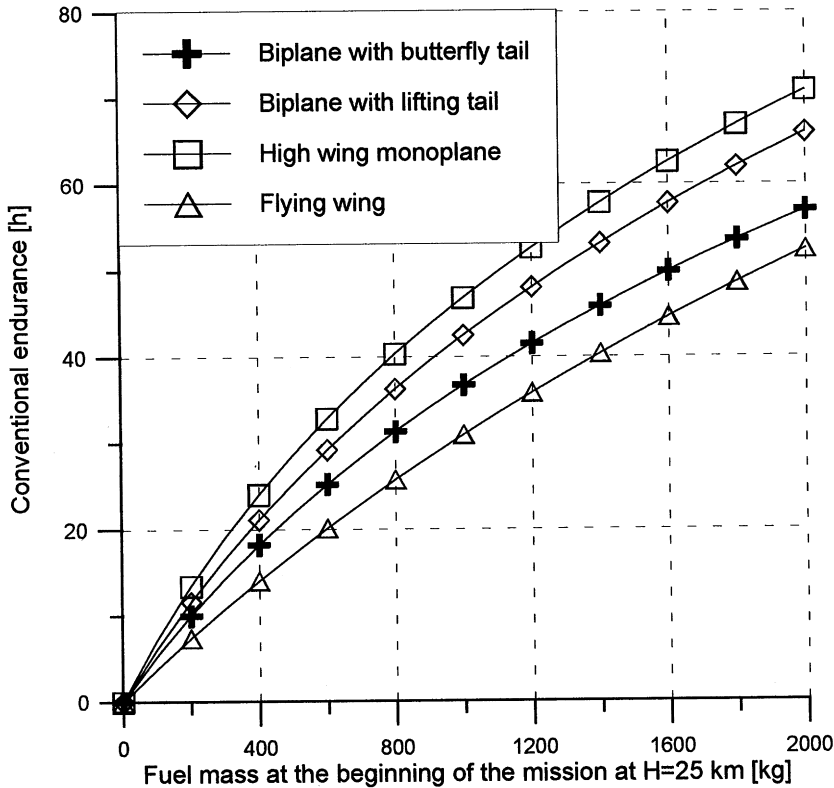


Fig. 12. Conventional endurance versus initial fuel mass at the beginning of altitude flight, computed for the maximum endurance flight conditions.

A comparison between the four aerodynamic configurations on Fig. 8 can also be made using the so-called conventional endurance and conventional range. These are endurance and range computed under the assumptions that the specific fuel consumption is equal to $0.3 \text{ kg}/(\text{kW}\cdot\text{h})$ and that the propeller efficiency is equal to 1. Conventional endurance and range are shown in Figs. 12 and 13 for all four configurations, versus the total fuel mass, being in aircraft tanks at the beginning of flight at the prescribed ceiling, assumed to be 25 km. Both range and endurance have been computed for maximum endurance flight conditions. Figs. 12 and 13 allow to rank the considered configurations (with respect to their maximum endurance for example) in the following order: (1) high-wing monoplane, (2) biplane with lifting horizontal tail, (3) biplane with butterfly tail and (4) flying wing. The difference in range between the high-wing monoplane and the biplane (Fig. 13) is huge—the maximum relative difference in range is about 36%. The difference in endurance is much smaller—about 6% only. The biplane uses almost the same fuel to reach the service ceiling but has a 290 higher operating empty mass than the high-wing monoplane. However, one should take into account that the capacity of the integral fuel tank and wing fuel compartments in the high-wing monoplane configuration is in reality limited to 1500 kg. In fact, the endurance and range at 25 km altitude for the high-wing monoplane configuration will be reduced and this is the reason why special attention was given to the biplane.

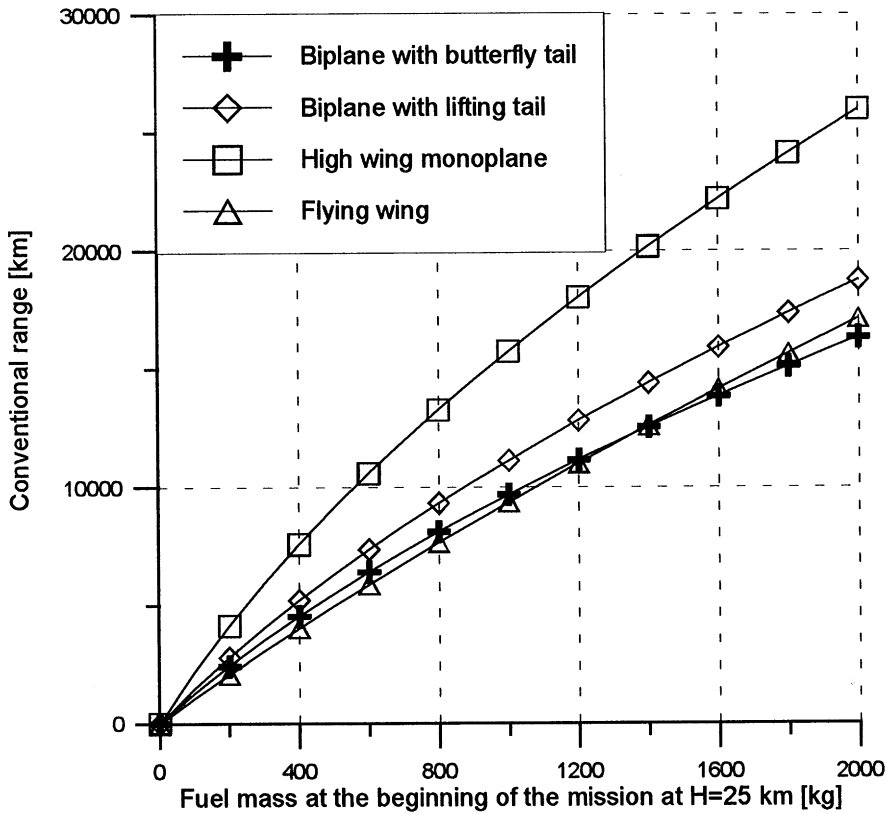


Fig. 13. Conventional range versus initial fuel mass at the beginning of altitude flight computed for the maximum endurance flight conditions.

The conventional endurance was computed using the following definition:

$$q_e = - \frac{dm}{P(m) dt}, \quad (5)$$

and by integration,

$$\text{endurance} = \frac{-1}{q_e} \int_{m_1}^{m_2} \frac{dm}{P(m)}, \quad (6)$$

where $P(m)$ denotes the required power (as function of aircraft mass); q_e the specific fuel consumption; and t the time; m_1, m_2 the initial and final total mass of the aircraft. Function $P(m)$ was taken from Fig. 11, it is dotted line corresponding to the economical speed where $(C_L)^{3/2}/C_D$ is a maximum.

The longest endurance for high-altitude flights is usually of primary importance. Such missions should be performed with the economical speed. However, if the longest range is more important than the endurance, the mission should be performed with the minimum drag and optimum speed

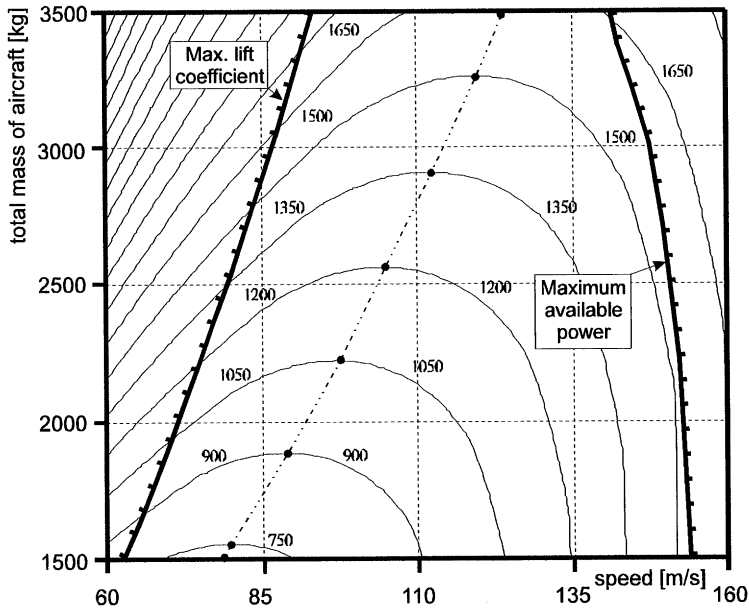


Fig. 14. Thrust (in N) required to fly at 25 km altitude for the biplane with lifting tail.

(note that the optimum speed changes as aircraft mass decreases). In such a case the conventional range can be computed from

$$q_e = - \frac{dm}{T(m) V(m) dt} \quad (7)$$

and by integration,

$$\text{range} = \frac{-1}{q_e} \int_{m_1}^{m_2} \frac{dm}{T(m)}, \quad (8)$$

where $T(m)$ denotes the required thrust (as function of aircraft mass) and $V(m)$ the optimum speed at the given aircraft mass. Function $T(m)$ was taken from Fig. 14, it is the dotted line corresponding to the optimum speed where (C_L/C_D) is a maximum.

Conventional ranges, computed for the maximum-range flight conditions, versus the total fuel mass being in aircraft tanks at the beginning of flight and at the prescribed ceiling (25 km), are shown in Fig. 15 for all four configurations. Comparing the ranges of corresponding configurations given in Fig. 15 (computed for the maximum range flight conditions) and in Fig. 13 (computed for the maximum endurance flight conditions) one can notice considerable differences. For example, the high-wing monoplane conventional range corresponding to the 2000 kg of initial fuel mass is equal to 26,000 km in Fig. 13 (for maximum endurance flight conditions) and 32,000 km in Fig. 15 (for maximum-range flight conditions), respectively.

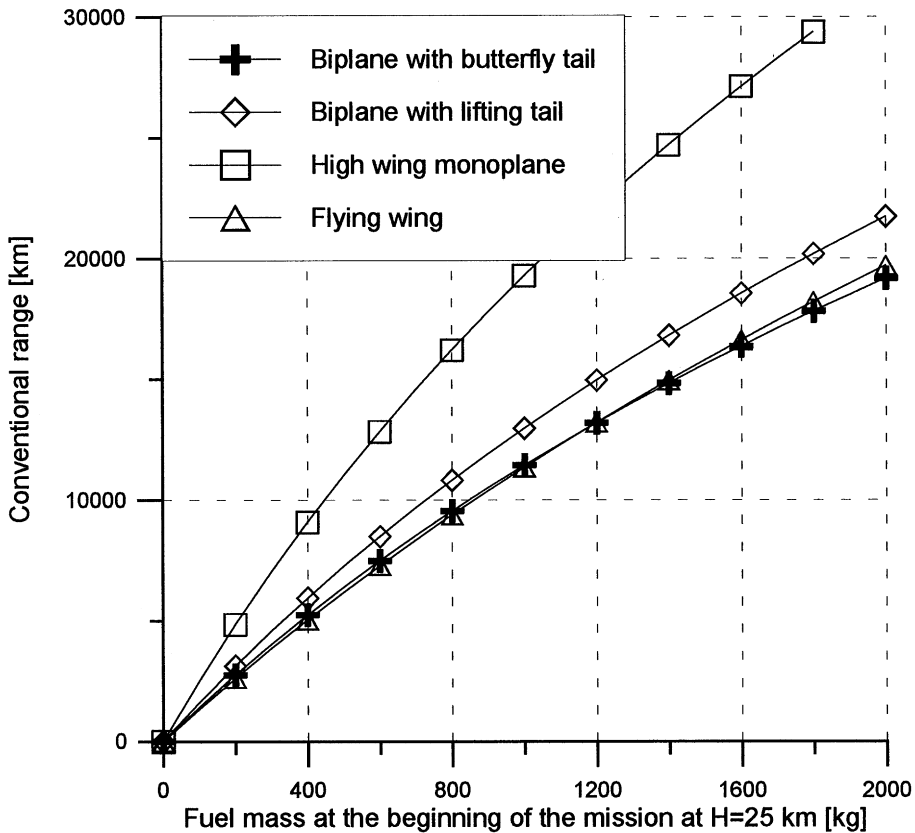


Fig. 15. Conventional range versus initial fuel mass at the beginning of altitude flight, computed for the maximum-range flight conditions.

It is known that pure flying wings tend to have higher trimmed L/D values compared with tailed configurations. With a conventional flight control system the flying wing can have dynamic handling deficiencies due to the low inherent pitch damping. However, with today's highly reliable digital flight control systems (fly-by-wire) these deficiencies can be compensated. The biplane with lifting tail can be attractive because of the smaller main wing lift and less induced drag. So, it has an increased total lift to drag ratio. However, usually it means reduced stability, sometimes even instability. The level of instability at aft and forward centre-of-gravity positions should be the subject of a detailed study of benefits in the so-called 'trimmed lift-to-drag ratio'. The configuration will need the longitudinal stability augmentation system which must generate a value of incremental static margin and artificial dynamic stability. The authors are conscious that selection of such unconventional configurations can be risky, first of all because of the lack of data and experience dealing with these configurations. However, it seems that dealing with an unconventional goal one needs an unconventional solution, and after detailed suitable analysis of such an unusual design concept one may have a better chance to fulfil the mission requirements.

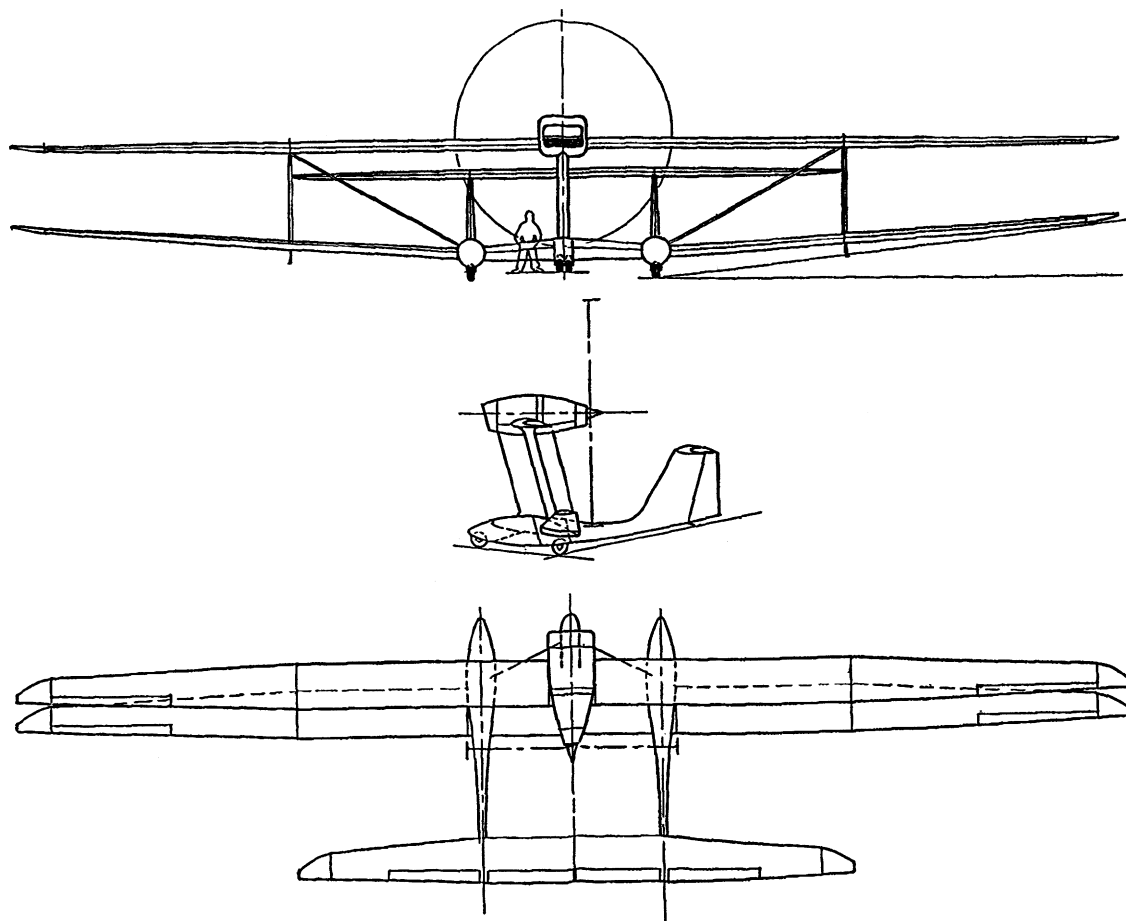


Fig. 16. Preliminary three-view elevation of lifting tail biplane.

6. Description of the biplane design

In this chapter some preliminary considerations about design layout and servicing are included. We limit our remarks to the lifting tail configuration only; a three-view elevation of this configuration is shown in Fig. 16. A tricycle landing gear including wing tip bumpers has been selected. The double front wheels are hydraulically steered. The single wheels of the main undercarriage can be hydraulically braked. It is expected that weight savings of over 30% can be achieved using composite materials instead of aluminium alloys. The wings are made of graphite and boron composite materials. All segments of the main wing have three-compartment boxes, with stressed upper and lower surface skins. Wings webs are made of graphite composite. Latticed ribs in the vicinity of the trailing edge are made of aluminium alloy. The aircraft can be easily taken to pieces (Fig. 17) for easy transportation and maintenance. The only difficulty expected in servicing is caused by placing the power-unit at high height, so a crane is needed in the case of big part replacement. Because the biggest aircraft parts do not exceed 8×1.5 m (in the case of wing) and

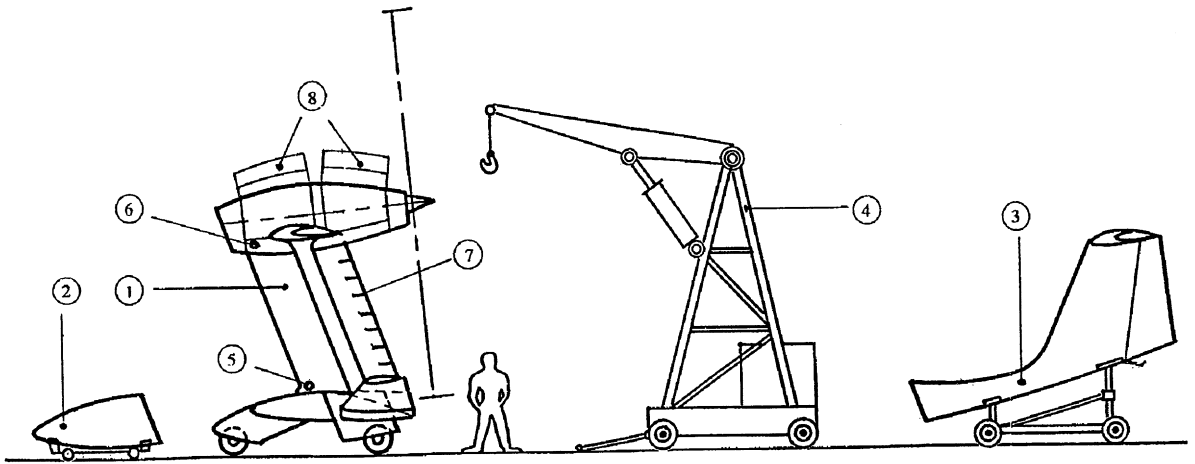


Fig. 17. Aircraft is easy for transportation and service. Legend: (1) aircraft main frame, (2) container with research equipment, (3) rear part—fuselage beam and empennage, (4) crane—static lifting capacity 500 kg, (5) pressure filler, (6) gravitational filler, (7) foot plates for engine maintenance, (8) engine covers.

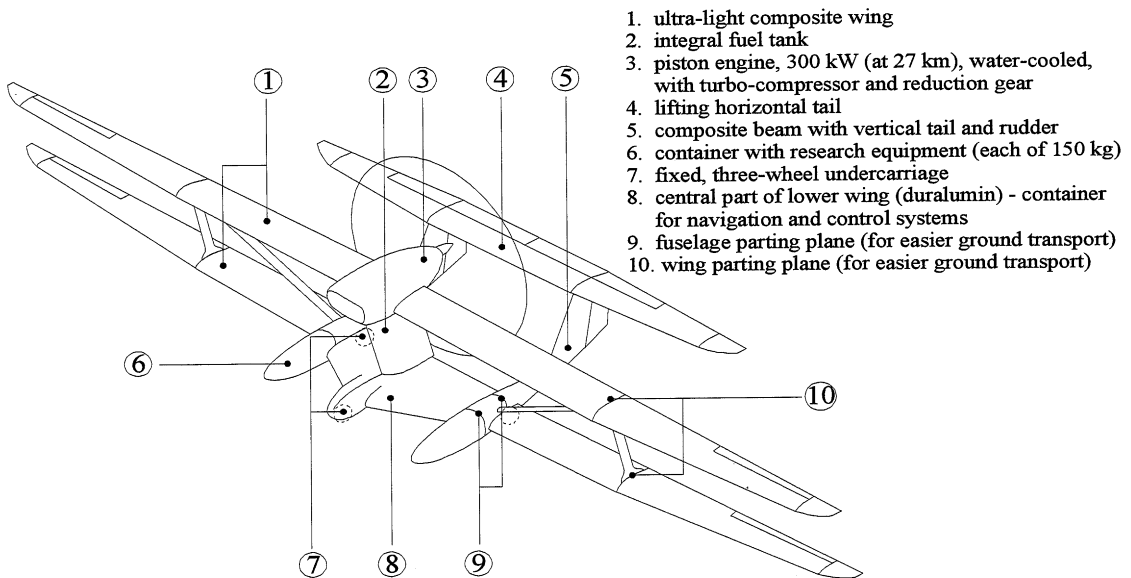


Fig. 18. Biplane with lifting tail—scheme of main design units.

7×2.6 m (in the case of centre-wing section), then transportation is easy and possible using standard containers. Flight will be guided using either an autopilot or a direct remote control module. The lifting surface control system is duplicated and managed by an on-board computer. In the case of any power-unit failure the aircraft can be brought down in gliding flight and safely landed. A special system of self-destruction in emergency is also built-in. The main parts of the whole design are shown and described in Fig. 18. Essential parts of the centre-wing section are shown in Fig. 19.

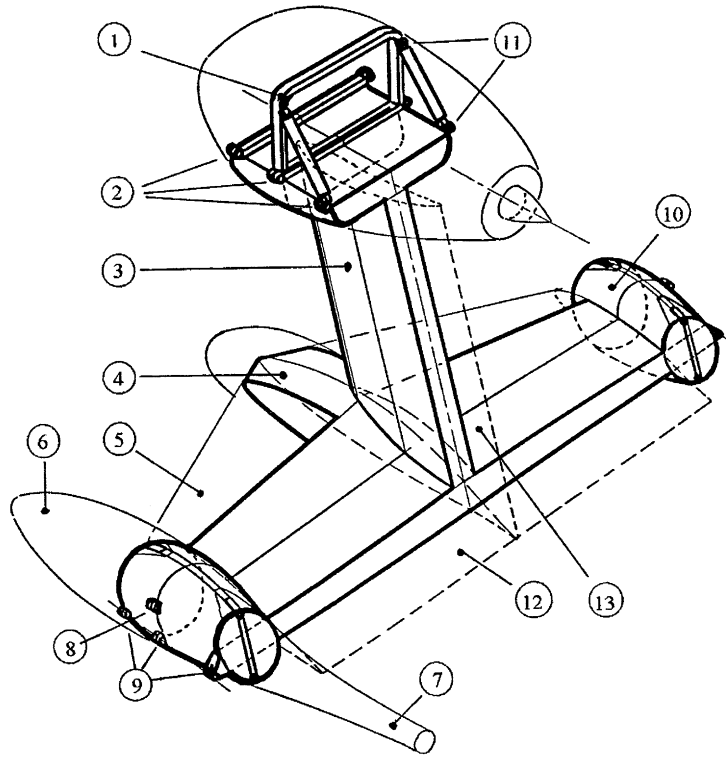


Fig. 19. Centre-wing section: (1) main frame of nacelle and power-unit, (2) upper wing attachment joints, (3) vertical pylon – fuel tank, (4) torsion box for the front wheel attachment, (5) cargo compartment, (6) container with research equipment (each of 150 kg), (7) fuselage beam, (8) wing strut attachment joints, (9) lower wing attachment joints, (10) nacelle of main undercarriage, (11) attachment joints for the engine mount with reduction gear, (12) accessories compartment, (13) luggage loading door.

7. Some remarks about flight dynamics

The wing loading changes from about 300 N/m^2 during take-off to about 150 N/m^2 during landing. Fig. 20 shows the change of wing loading for the biplane with lifting tail configuration under the assumption that the main mission takes place at 25 km height. As it was mentioned in Section 4, flying at constant altitude does not depend on trim changes – the reduced Mach number depends only on wing loading and flight altitude and does not depend on speed, lifting force coefficient and elevator deflection. Burning of the fuel decreases the wing loading and decreases the reduced Mach number. However, if one assumes that the aircraft mass is constant (hence the wing loading is constant) then the reduced Mach number $M = Ma \sqrt{C_L}$ has to be a constant too (of course it means that a number of choices are possible—smaller Mach number with greater lift coefficient or vice versa—greater Mach number at lower lift coefficient). Wing loadings for ‘Harve Aircraft’, Condor and Theseus as a linear function of the squared reduced Mach number are shown in Fig. 20. It should be emphasised that the very low wing loading, especially just before landing, can involve some problems with gust.

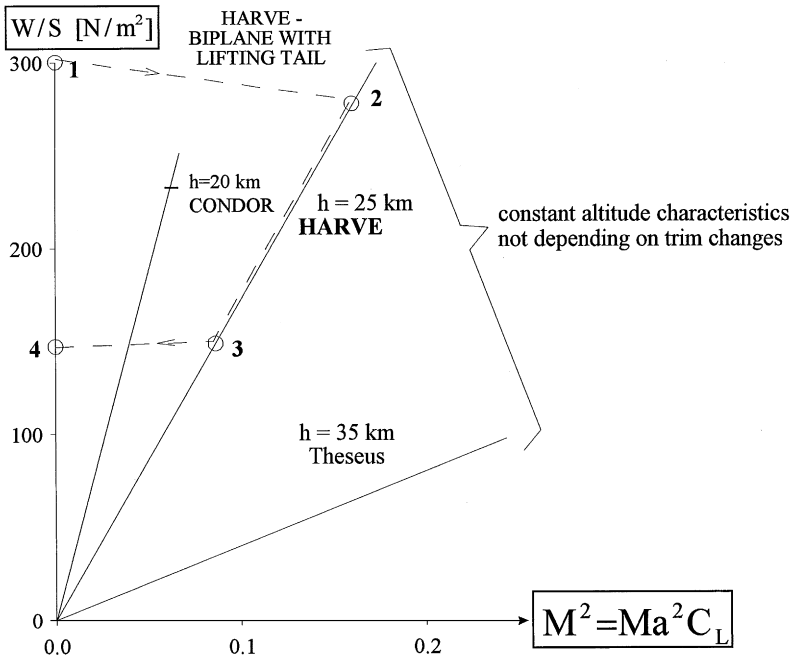


Fig. 20. Change of wing loading and square of reduced Mach number along flight trajectory: (1) take-off, (2) beginning of patrol mission, (3) beginning of descent, (4) landing.

The analysis of the damping and frequency for the natural modes of vibration has been made using STB package [26]. Eigenvalues of a state matrix of 8*8 order have been computed and some of them are shown in Figs. 21 and 22. From these figures it can be concluded that the phugoid mode, initially unstable at lower altitudes and smaller flight speeds, became stable for heights greater than 15 km. Similarly the spiral mode, unstable in the medium speed range at Sea Level stabilises if the altitude is greater than 5 km. Damping of the Dutch roll degrades somewhat when the altitude increases. It is clear that from the stability point of view the worst case is the flight at Sea Level.

8. Conclusions and recommendations

One can expect that the overall progress in the aeronautical technology will soon enable that unmanned non-expensive aircraft, with airbreathing and propeller driven engines, could be operated at the ultra high altitude. Selecting the power unit, gear box, compressors, coolers and first of all the aerodynamic configuration is the key point to fulfil this engineering task successfully. Flying at high, stratospheric altitudes leads to rather unusual Mach and Reynolds number combination—low Reynolds numbers and sometimes transonic speeds. A flying wing configuration does not seem to be a good choice. Its wing span (50 m) is very high in comparison to

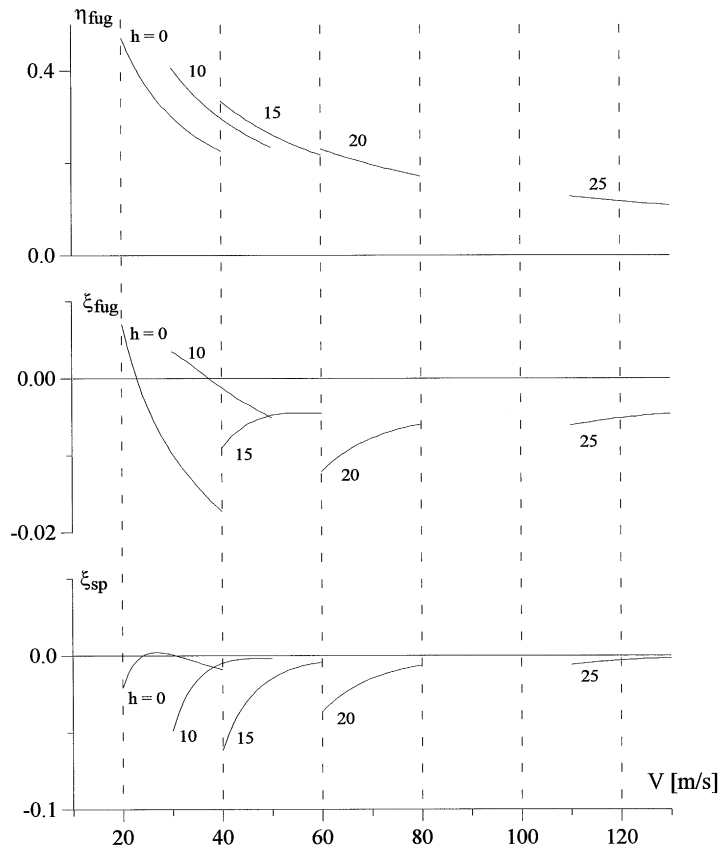


Fig. 21. Damping and frequency coefficients of the phugoid and spiral modes versus speed for different flight altitudes.

other configurations. The flying wing configuration does not have any external fuel tank and its zero-lift coefficient (0.010) is the lowest of the all considered configurations. However, the whole fuel has to be placed inside the wing increasing its weight and area. In result the wing aspect ratio is lower than for high-wing monoplane and is the same order as for the biplane. At lower speed the performances depend first of all on induced drag (not on the zero-lift coefficient) and it is the explanation why the flying wing is less effective for long endurance patrol mission than other considered configurations. Increasing the flying wing aspect ratio (keeping the wing area constant) is not possible because the operational limits (50 m span is high enough) and expected aeroelastic problems. Also, the aspect ratio is strongly limited because it involves an excessive increase of structure mass and in consequence, decreases the endurance and range. Because the propeller diameter is usually very large (about 7 m), then for a flying wing configuration in order to keep sufficient clearance during take-off and landing one has to design the power-unit of variable angle of inclination. A biplane with lifting tail seems to be a good choice for long endurance and high payload flights, but only in the case when loiter speeds are below the compressible range. All presented results were obtained under the assumption that local Reynolds numbers are

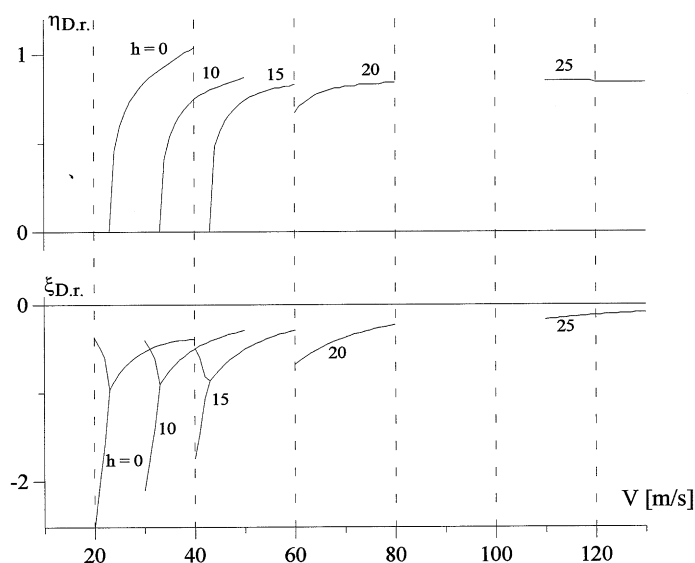


Fig. 22. Damping and frequency coefficients of the Dutch roll versus speed for different flight altitudes.

above-critical. Analysis shows that dynamic stability of biplane improves if the flight altitude increases—the phugoid and spiral mode stabilise for higher altitudes. Investigation shows that from the dynamic stability point of view the worst case corresponds to the flight at Sea Level. Unfortunately, due to limited length of this paper some very important engineering problems (for example susceptibility to gusts, flutter-resistance, divergence of control surfaces) has not been analysed here and should be addressed in the future.

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